Omar Alhalawani

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Education

Bachelor of Engineering in Computer Systems Engineering — Ottawa, ON
Carleton University — CGPA: 10.80/12 — *2024 – 2029*

- Ontario Professional Engineers Scholarship — Dean's Honour List

Technical Skills

RTL & Digital Design: SystemVerilog, Verilog, FPGA development, synchronous design, FSMs, datapaths
Verification: UVM, PyUVM, cocotb, self-checking testbenches, regression testing, waveform analysis
Programming & Automation: Python, Java, Bash, Makefiles, C, EDA flow automation
Tools & Workflow: Linux, Git, Vivado, Jira, Confluence, Agile/Scrum
Professional Training: Doulos Comprehensive SystemVerilog Online, July 2026

Experience

ASIC Engineer I Co-op – Digital Verification Methodology — May 2026 – August 2026
Ciena — *Ottawa, ON*

- Evaluated PyUVM and cocotb for AI-assisted verification, generating and validating self-checking testbenches from RTL specifications and verification plans.
- Designed fault-injection and performance experiments, analyzed simulation and regression results, and presented verification methodology recommendations to Ciena engineers and management.
- Developed Python scripts to analyze Makefile-based ASIC verification compile flows and explore methods for reducing unnecessary recompilation.

Embedded Hardware & Research Instrumentation Assistant — Sept. 2025 – April 2026
Carleton University – Engineering Laboratories — *Ottawa, ON*

- Debugged timing, signal-integrity, and hardware/software integration issues in experimental instrumentation.
- Developed firmware and automation for real-time data acquisition systems using microcontrollers, sensors, and ADCs.

Teaching Assistant – Programming and Digital Systems — Sept. 2025 – May 2026
Carleton University – Department of Systems & Computer Engineering — *Ottawa, ON*

- Mentored students in synchronous RTL design, FSMs, simulation debugging, waveform analysis, and HDL validation.

Projects

Synchronous FIFO RTL Design & Verification — *SystemVerilog*

- Designed a synthesizable parameterized FIFO with pointer logic, full/empty detection, and synchronous handshaking.
- Developed a self-checking SystemVerilog testbench with directed and corner-case tests and automated validation.

JPacman Software Engineering Project — *Java, Git, Jira, Confluence, Agile/Scrum*

- Leading a three-person team through six two-week Agile sprints on an open-source Java Pac-Man implementation, coordinating work across movement, collisions, scoring, and ghost behaviour.
- Using Jira, Confluence, and Git to manage requirements, design, implementation, testing, and verification throughout the development process.

LumaLink – Real-Time Optical Communication System — *QNX RTOS, C, Python, Raspberry Pi*

- Developed the C receiver using the QNX Camera API to capture and decode high-speed optical signals through a Raspberry Pi camera.
- Integrated the receiver with a Python-based transmitter and processing pipeline under time-critical frame-processing constraints.

Extracurricular Involvement

Biomedical Carleton Applied Research and Engineering (BioCARE) — *Sept. 2025 – Present*

- Developed an EMG acquisition system using MyoWare sensors, Arduino/C++, and a Python/Linux processing pipeline.